

National University of Computer and Emerging Sciences



Software for Mobile Devices

Spring 2025

Assignment 3

Objective:

To develop a multi-tab mobile application that allows users to schedule tasks, view past activities, manage notifications, and personalize their profile with dark/light mode using **SQLite** and **SharedPreferences** only.

Assignment Requirements:

Your application must consist of a **TabLayout with ViewPager** containing four screens:

1. **Schedule Tab**
2. **Past Tab**
3. **Notification Tab**
4. **Profile Tab**

Schedule Screen:

- **Functionality:**
 - Add a new task (title, description, date & time).
 - View a list of upcoming tasks (only those with future dates).
- **Storage:**
 - Use **SQLite** to store tasks with fields: id, title, description, datetime, status.

Past Screen:

- **Functionality:**
 - Automatically display tasks whose date/time is **in the past**.
 - Read from the same **SQLite** table as Schedule.
- **Sorting:**
 - Show latest past tasks at the top.

Notification Screen:

- **Functionality:**
 - Display a list of dummy notifications.
 - You can hardcode or simulate the addition of new notifications.
- **Storage:**
 - Save notifications in **SQLite** with fields: id, message, datetime.

Profile Screen:

- **Functionality:**
 - Edit user's name and email.
 - Select Dark or Light Mode (theme switch).
- **Storage:**
 - Save user info and theme preference using **SharedPreferences**.
- **Theme Implementation:**
 - On app start, check the theme from SharedPreferences and apply accordingly.

Screens:

